



## What's Hot about MongoDB, HBase and Cassandra

MongoDB, HBase and Cassandra are the buzz words in the database domain. All three have their strong points and their failings. The author gives a very brief overview of each one, and prompts readers to explore these databases further.

In the autumn of 1998, Eric Brewer, a scientist at the University of California, Berkeley, presented a conjecture that fundamentally altered the perception of distributed data storage systems. Proved formally by researchers from MIT two years later, it is today known as the CAP Theorem, and is one of the factors that separates SQL from NoSQL.

### So what exactly is this theorem?

We are all aware that there are fundamental principles that govern database systems, namely, the ACID properties—atomicity, consistency, isolation and durability. Brewer speculated that a distributed system can only provide two out of the three—consistency, availability and partition tolerance. Of course, this is a different ‘consistency’ from the ACID properties, in that the CAP theorem defines consistency as ‘every read receiving the most recent write’ while ACID treats it as a ‘valid state for the system to be in’.

While SQL, being a relational database, deals primarily with ‘structured’ or schema-based data, NoSQL allows for the manipulation of unpredictable and ‘unstructured’ information in an attempt to fulfil the requirements of many of the Big Data models and architectures being integrated

into businesses today. SQL is often criticised by many for its perceived ‘inability to scale’; while it can scale vertically, this also requires considerable hardware and, in turn, investment. Specifically, the scalability issue arises with latency and inadequate failover capabilities that are addressed often by advocating a switch to NoSQL based technologies, short for ‘Not only SQL’.

### NoSQL to the rescue

Dare Obasanjo intuitively explains the difference between SQL and NoSQL in a blog post on this subject. He claims that SQL is similar to a vehicle with automatic transmission, while NoSQL falls into the manual transmission category. While automatic vehicles enforce certain integrity checks and rules that internally define the system, with manual vehicles, a lot of this responsibility falls on the users who may choose to forego some constraint checks and define their own instead, resulting in a performance improvement.

The reality, however, is that not everyone is required to use NoSQL as their *de facto* standard for performance because, as Obasanjo claims, even a manual car in traffic conditions is bound to perform similar to an automatic